

Limit laws

$$\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$\lim_{x \rightarrow a} [c f(x)] = c \lim_{x \rightarrow a} f(x)$$

$$\lim_{x \rightarrow a} [f(x) g(x)] = \lim_{x \rightarrow a} f(x) \lim_{x \rightarrow a} g(x)$$

$$\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad \text{if } \lim_{x \rightarrow a} g(x) \neq 0$$

$$\lim_{x \rightarrow a} c = c$$

$$\lim_{x \rightarrow a} x = a$$

Standard limits

$$\lim_{x \rightarrow \infty} \frac{1}{x^p} = 0 \quad (p > 0)$$

$$\lim_{x \rightarrow \infty} x^r = 0 \quad (0 < r < 1)$$

Limit laws

$$\lim_{n \rightarrow \infty} [a_n + b_n] = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n$$

$$\lim_{n \rightarrow \infty} [c a_n] = c \lim_{n \rightarrow \infty} a_n$$

$$\lim_{n \rightarrow \infty} [a_n b_n] = \lim_{n \rightarrow \infty} a_n \lim_{n \rightarrow \infty} b_n$$

$$\lim_{n \rightarrow \infty} \left[\frac{a_n}{b_n} \right] = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}$$

$$\lim_{n \rightarrow \infty} c = c$$

Standard limit

$$\#9 \lim_{n \rightarrow \infty} \arctan(c_n) = \frac{\pi}{2} \quad (c > 0)$$

Sandwich theorem

If $g(x) \leq f(x) \leq h(x)$, and
 $\lim_{x \rightarrow \infty} g(x) = \lim_{x \rightarrow \infty} h(x) = L$
 then, $\lim_{x \rightarrow \infty} f(x) = L$

Example

Evaluate $\lim_{x \rightarrow 0} [x^2 \sin(\frac{1}{x})]$

$$-1 \leq \sin(\frac{1}{x}) \leq 1, x \neq 0$$

$$-x^2 \leq x^2 \sin(\frac{1}{x}) \leq x^2$$

$$\lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} h(x) = 0$$

$$\Rightarrow \lim_{x \rightarrow 0} x^2 \sin(\frac{1}{x}) = 0$$

[sandwich theorem]

Standard Limit

$$\#1 \lim_{n \rightarrow \infty} \frac{1}{n^p} = 0 \quad (p > 0)$$

$$\#2 \lim_{n \rightarrow \infty} r^n = 0 \quad (|r| < 1)$$

$$\#3 \lim_{n \rightarrow \infty} a^{\frac{1}{n}} = 1 \quad (a > 0)$$

$$\#4 \lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$$

$$\#5 \lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0 \quad (a \in \mathbb{R})$$

$$\#6 \lim_{n \rightarrow \infty} \frac{\log(n)}{n^p} = 0 \quad (p > 0)$$

$$\#7 \lim_{n \rightarrow \infty} (1 + \frac{a}{n})^n = e^a \quad (a \in \mathbb{R})$$

$$\#8 \lim_{n \rightarrow \infty} \frac{n^p}{a^n} = 0 \quad (p \in \mathbb{R}, a > 1)$$

Continuity

Below functions are continuous everywhere:

- ▷ polynomials
- ▷ trigonometric functions
- ▷ exponential functions
- ▷ logarithm functions
- ▷ nth root functions $\sqrt[n]{x}$
- ▷ absolute value function $|x|$
- ▷ hyperbolic functions

if f and g are continuous then $f+g, cf, f \cdot g, \frac{f}{g}$ and $g(f(x))$ also continuous
 e.g. $\sin(2 \log(x))$ is continuous

Partial sums

$$S_1 = a_1$$

$$S_2 = a_1 + a_2$$

$$S_n = a_1 + \dots + a_n$$

Geometric Series

$$\sum_{n=0}^{\infty} ar^n = \sum_{n=1}^{\infty} ar^{n-1}$$

The series diverge if $|r| \geq 1$
 converge if $|r| < 1$
 and $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$

Continuity for limits of composition

if $\lim_{x \rightarrow a} g(x) = b$ and f is continuous at b , then
 $\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x)) = f(b)$

* also holds for $x \rightarrow \infty$ and for sequence

L'Hôpital's Rule

★ Indeterminate form $\frac{0}{0}, \frac{\infty}{\infty}$
 ★ $f(x)$ and $g(x)$ are differentiable near $x=a$

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

★ 如果用在 sequence 要 convert

Harmonic p series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

The series diverge if $p \leq 1$
 converge if $p > 1$
 → converge to 0

Properties of Series

$$\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$$

converge converge converge
 大家一起 converge 吧

< Divergence Test >

If $\lim_{n \rightarrow \infty} a_n \neq 0$
 THEN $\sum_{n=1}^{\infty} a_n$ diverges

等于0代表不了任何东西

< Ratio Test >

Step 1: $a_n > 0$
 Step 2: $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L$
 The series converge if $L < 1$
 diverge if $L > 1$
 inconclusive if $L = 1$

< Comparison Test >

Step 1: $a_n > 0 \rightarrow$ try comparison test
 roughly as $n \rightarrow \infty$

IF a_n converge, # upper bound

$$\text{例 } a_n = \frac{3+5n}{2n^2+n+2} \leq \frac{3+5}{2n^2} = \frac{4}{n^2}$$

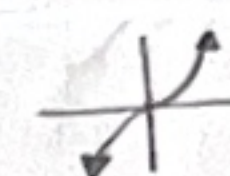
$$n \geq 1$$

IF a_n diverge, # lower bound

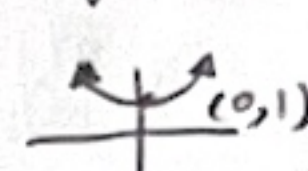
$$\text{例 } \frac{n^2+4}{n^3+5} \geq \frac{n^2}{n^3+5n^3} = \frac{n^2}{6n^3} = \frac{1}{6n}$$

【Harmonic p series】

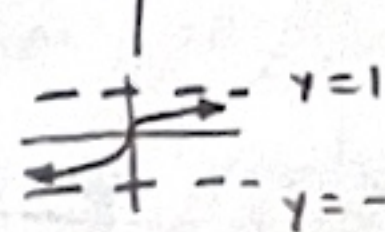
sinh(x)



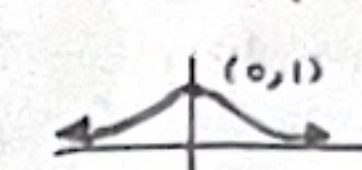
cosh(x)



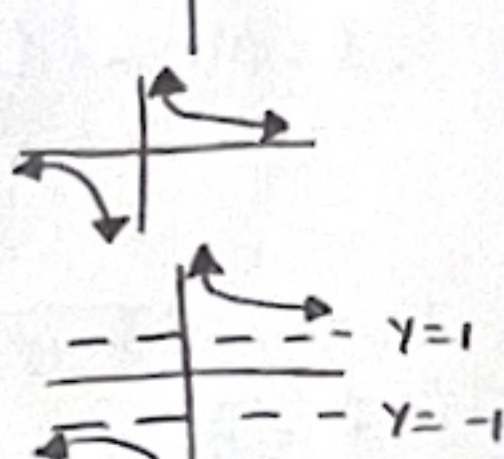
tanh(x)



sech(x)



coth(x)



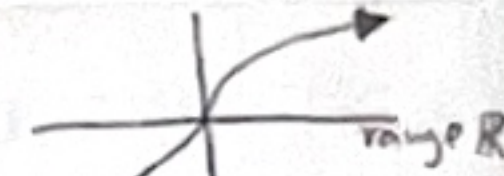
arcsech(x)

$0 < x \leq 1$

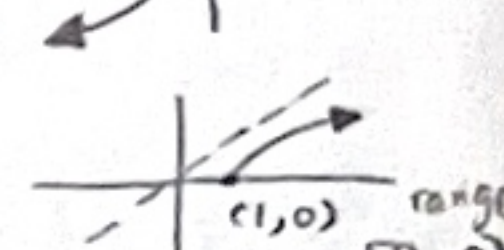
arcroth(x)

$x < -1$ or $x > 1$

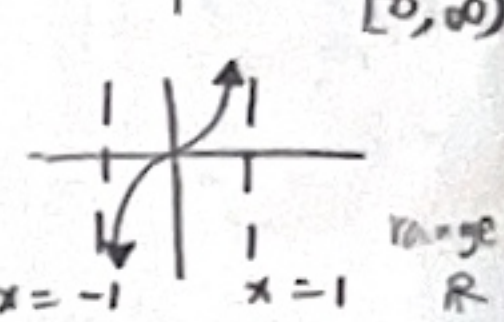
arcsinh(x)



arc(cosh(x))



arctanh(x)



$$\frac{d}{dx} [\text{sech}(x)] = -\text{sech}(x) \tanh(x)$$

$$\frac{d}{dx} [\text{cosech}(x)] = -\text{cosech}(x) \coth(x)$$

$$\frac{d}{dx} [\coth(x)] = -\text{cosech}^2(x)$$

$$\frac{d}{dx} [\text{arsinh}(x)] = \frac{1}{\sqrt{x^2+1}} \quad | x \in \mathbb{R}$$

$$\frac{d}{dx} [\text{arcosh}(x)] = \frac{1}{\sqrt{x^2-1}} \quad | x > 1$$

$$\frac{d}{dx} [\text{arctanh}(x)] = \frac{1}{1-x^2} \quad | -1 < x < 1$$

Partial Fraction

$$(x-a) \mid \frac{A}{x-a}$$

$$(x-a)^r \mid \frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \dots + \frac{A_r}{(x-a)^r}$$

$$(x^2+bx+c) \mid \frac{Ax+B}{x^2+bx+c}$$

$$(x^2+bx+c)^r \mid \frac{Ax+B}{x^2+bx+c} + \frac{A_2x+B_2}{x^2+bx+c} + \dots$$

First order ODE

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Integration factor

$$I(x) = e^{\int P(x) dx}$$

Products and Division

$$z \cdot w = r_1 r_2 e^{i(\theta + \phi)}$$

$$\frac{z}{w} = \frac{r_1}{r_2} e^{i(\theta - \phi)}$$

Powers

$$z^n = r^n e^{i \cdot n \cdot \theta}$$

Diff via Complex Exponential

$$\frac{d}{dt} (e^{-t} \cos(t)) = 2^{20} e^{-t} \cos(t)$$

$$\rightarrow e^{-t} \cos(t) = e^{-t} \cdot \text{Re}(e^{it})$$

$$= \text{Re}(e^{(-1+i)t})$$

$$\rightarrow \text{Re}(\frac{d}{dt} e^{(-1+i)t}) = \text{Re}((-1+i)e^{(-1+i)t})$$

$$\rightarrow \text{de Moivre's theorem } (1+i)^{40} = 2^{20} e^{20i}$$

Integrate via Complex Exponential

$$\textcircled{1} \int e^{3x} \sin(2x) dx = e^{3x} \text{Im}(e^{2ix})$$

$$= \text{Im}(e^{(3+2i)x})$$

$$\textcircled{2} \int e^{3x} \sin(2x) dx = \text{Im}(e^{(3+2i)x}) dx$$

$$= \text{Im}[\int e^{(3+2i)x} dx]$$

$$= \text{Im}[\frac{1}{3+2i} e^{(3+2i)x} + c + di]$$

$$= \text{Im}[\frac{1}{3+2i} \cdot \frac{3-2i}{3-2i} e^{3x} \cdot e^{2ix} + c + di]$$

$$= \frac{e^{3x}}{13} (-2 \cos(2x) + 3 \sin(2x))$$

$$+ d, d \in \mathbb{R}$$

Integration by parts

$$\int u \cdot \frac{dv}{dx} dx = u \cdot v - \int v \cdot \frac{du}{dx} dx$$

Integration by substitution

$$\sqrt{a^2-x^2}, \sqrt{a^2+x^2} \quad \begin{cases} x = a \sin(\theta) \\ \text{OR } x = a \cos(\theta) \end{cases}$$

$$\sqrt{a^2-x^2}, \sqrt{a^2+x^2} \quad x = a \sinh(\theta)$$

$$\sqrt{x^2-a^2}, \sqrt{x^2+a^2} \quad x = a \cosh(\theta)$$

$$\frac{1}{a^2+x^2}, \frac{1}{(a^2+x^2)^2} \quad x = a \tan(\theta)$$

$$\frac{1}{a^2-x^2} \quad x = \tanh(\theta)$$

$$x = 5 \cosh(\theta), \theta = \text{arcosh}(\frac{x}{5}) \geq 0$$

First order ODE

Method 1: $\frac{dy}{dx} = f(\frac{y}{x})$

step 1: $a = \frac{y}{x}$

$y = x \cdot u$

$\frac{dy}{dx} = x \cdot \frac{du}{dx} + u \cdot 1$

step 2: 两边相减

$x \frac{du}{dx} + u = \frac{y}{x} + \cos^2(\frac{y}{x})$

$= u + \cos^2(u)$

Method 2: $\frac{dy}{dx} + P(x)y = Q(x)y^n$

step 1: $u = y^{1-n}$

step 2: 然后求 $\frac{du}{dx}$ 再套

given $\frac{dy}{dx}$

和 $\frac{du}{dx} = -2y^{-3} \frac{dy}{dx}$

Equation of Motion

$m\ddot{y} + \beta\dot{y} + ky = 0$

Net force = $m\ddot{y}$

$= mg - k(s+y) - \beta\dot{y}$

characteristic equation

$m\lambda^2 + \beta\lambda + k = 0$

$\lambda = \frac{-\beta \pm \sqrt{\beta^2 - 4mk}}{2m}$

no damping, $\beta = 0$, $\lambda = \pm ib$

underdamp, $0 < \beta < 2\sqrt{mk}$, $\lambda = a \pm bi$

critical damp, $\beta = 2\sqrt{mk}$, $\lambda = a, a$

overdamp, $\beta > 2\sqrt{mk}$, $\lambda = a, b$

★ SECOND DERIVATIVES ★

Chain rule

$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$

$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}$

$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$

Directional Derivatives

Gradient Vector

$\nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$

PARTIAL INTEGRATION

$\int 3x^2 y dx$

$= [\frac{3}{3} x^3 y] + C(y)$

$= x^3 y + C(y)$

Find volume for 1-x

domain $R \in \mathbb{R}^2$

is $R = [0, 1] \times [0, 2]$

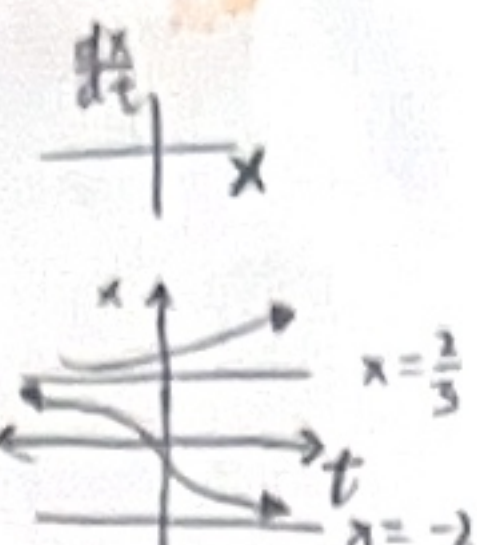
$0 \leq x \leq 1$ and $0 \leq y \leq 2$

volume = $\int_0^2 \int_0^1 1-x dx dy$

is match

Phase plot

Family of solutions



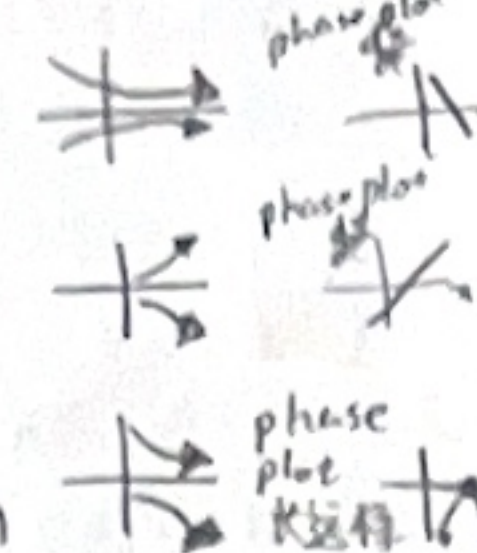
when $x > \frac{2}{3}$, $\frac{dx}{dt} > 0$

when $x < \frac{2}{3}$, $\frac{dx}{dt} < 0$

stable equilibrium

unstable equilibrium

semistable equilibrium



例题

step 1 Find k:

At equilibrium: $mg = kx$

$k = \frac{mg}{x}$

$= 40 \text{ N}$

step 2

Establish equation of motion

Net force = $m \cdot \ddot{y}$

Net force = $mg - kx - \beta\dot{y}$

Resonance: $f(t)$ has the same form as one of the terms in the $g(s)$

例 G(s) = $A \cos(\omega t) + B \sin(\omega t)$

Directional Derivatives

$D_{\hat{u}} f|_{P_0} = (\frac{\partial f}{\partial x}|_{P_0}, \frac{\partial f}{\partial y}|_{P_0}) \cdot (u, v)$

dot product

Find direction vector

第一: in direction from (2,0)

towards $(\frac{1}{2}, 2)$

$\hat{u} = PQ = \vec{OQ} - \vec{OP}$

$= (\frac{1}{2}, 2) - (2, 0)$

$= (-\frac{3}{2}, 2)$

$\|\hat{u}\| = \sqrt{(-\frac{3}{2})^2 + 2^2} = \frac{5}{2}$

$\hat{u} = \frac{\hat{u}}{\|\hat{u}\|} = \frac{2}{5} (-\frac{3}{2}, 2)$

第二: in direction $\frac{\pi}{4}$ anticlockwise from positive x-axis

$\hat{u} = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

$\hat{u} = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

2022 S2 EXAM Q4V

Show $\text{arcsech}(x)$

$= \text{arctanh}(\sqrt{1-x^2})$

let $y = \text{arcsech}(x)$

$\text{sech}(y) = x$

$1 - \tanh^2(y) = \text{sech}^2(y)$

$\text{sech}^2(y) = x^2$

$1 - \tanh^2(y) = x^2$

$\tanh(y) = \sqrt{1-x^2}$

$y = \text{arctanh}(\sqrt{1-x^2})$

$y = \text{arctanh}(\sqrt{1-x^2})$

Flow rate problem

$\frac{dx}{dt} = \text{rate in} - \text{rate out}$

rate in = concentration coming in $\times$ volume coming in

rate out = concentration going out $\times$ volume going out

$= \frac{x}{100-t} \times 10$

Transient terms: terms decaying to 0 as $t \rightarrow \infty$ E.g. $\frac{1}{t} e^{-t}$

Steady state terms: terms not decaying to 0 as $t \rightarrow \infty$ E.g. 2

画 3D 图教程

$z = \sqrt{4-x^2+y^2}$

(-) Find level curve

let $z = c \geq 0$

$c^2 = 4-x^2+y^2$

选 $c=0$, $c=1$, $c=4$

$c=0, 0=4-x^2+y^2$ point (0,0)

$c=1, 1=4-x^2+y^2$

$c=4, 4=4-x^2+y^2$

(-) Find cross section

x-z plane $\rightarrow y=0$

y-z plane $\rightarrow x=0$

(-) $z \rightarrow y$ 标 function!

Direction of fastest increase or steepest

例题

In what direction does $f(x,y)$

$= x e^y$ (a) increase (b) decrease

most rapidly at (2,0)?

$\nabla f|_{(2,0)} = \hat{i} + 2\hat{j} = (1, 2)$

$\|\nabla f|_{(2,0)}\| = \sqrt{1^2 + 2^2} = \sqrt{5}$

(a) direction of fastest increase

is $\hat{u} = \frac{1}{\sqrt{5}} (\hat{i} + 2\hat{j})$

$= (\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}})$

(b) direction of fastest decrease

is $-\hat{u} = (-\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}})$

$-\hat{u} = (-\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}})$

$\frac{d}{dx} [\log_e(ax+tb)] = \frac{a}{ax+tb}$

$\int \frac{1}{ax+tb} dx = \frac{1}{a} \log_e(ax+tb)$

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4S (Homogeneous ODE)

$a\lambda^2 + b\lambda + c = 0$

① 2 real roots $\lambda = \lambda_1, \lambda_2$ $y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$

② 1 real root $\lambda = \lambda_1$ $y = Ae^{\lambda_1 x} + Bx e^{\lambda_1 x}$

③ 2 complex roots $\lambda = \alpha \pm \beta i$

$y = Ae^{\alpha x} \cos(\beta x) + Be^{\alpha x} \sin(\beta x)$

Particular (I.H.)

$= e^{3x} + 2x + 6$

① try $y_p(x) = ae^{bx} + c$

Tangent Plane

$z - z_0 = \frac{\partial f}{\partial x}|_{(x_0, y_0)} (x - x_0)$

$+ \frac{\partial f}{\partial y}|_{(x_0, y_0)} (y - y_0)$

求 $\frac{\partial f}{\partial x}$ 需先 rearrange equation

Approximate change

The approxi. change in f as

(x, y) change from (x_0, y_0)

to (x, y)

$\Delta f = \frac{\partial f}{\partial x}|_{(x_0, y_0)} (x - x_0)$

$+ \frac{\partial f}{\partial y}|_{(x_0, y_0)} (y - y_0)$

Stationary Points

Hessian function:

$H(x, y) = f_{xx} f_{yy} - (f_{xy})^2$

① $H(x_0, y_0) > 0$ and

$f_{xx}(x_0, y_0) > 0$

$\Rightarrow$ local minimum

② $H(x_0, y_0) > 0$ and

$f_{xx}(x_0, y_0) < 0$

$\Rightarrow$ local maximum

③ $H(x_0, y_0) < 0$

$\Rightarrow$ saddle point

④ $H(x_0, y_0) = 0$, test is inconclusive

$(\frac{x}{n+1})^n = (\frac{x}{n+1})^n$

$= (\frac{x(1+\frac{1}{n})}{n+1})^n$

$= \frac{1}{(1+\frac{1}{n})^n}$

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